Smallest multipliers with leading digit one

and a conjecture of Fröhlich (OEIS A190302)

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November 14, 2026

Abstract

Let $a(n)$ be the smallest $h \geq 1$ such that the decimal expansion of $n \cdot h$ begins with the digit 1 (OEIS A190302). A direct interval argument gives the exact formula: writing $10^d \leq n < 10^{d+1}$, one has $a(n) = 1$ when $n < 2 \cdot 10^d$, and $a(n) = \lceil 10^{d+1}/n \rceil$ otherwise; in the second case the value lies in $\{2, 3, 4, 5\}$. In particular $a(n) \leq 5$ for every $n \geq 1$, which settles the conjecture recorded on the entry by Felix Fröhlich (Jul 28 2018: "$a(n) < 6$ for all n", verified there up to $n = 10022141$). An equivalent bound – A187285$(n) \in \{n, 2n, 3n, 4n, 5n\}$, i.e. $a(n) \leq 5$ via A187285$(n) = n \cdot a(n)$ – was asserted without proof by C. R. Greathouse IV (2011) in the comments of OEIS A187285. The theorem is verified in Lean 4 with mathlib: the kernel-checked proof uses only the three standard axioms (propext, Classical.choice, Quot.sound), no native_decide and no sorry.

1 Statement of results

For $m \geq 1$, "the decimal expansion of m starts with 1" means $m \in [10^k, 2 \cdot 10^k)$ for some $k \geq 0$ (writing $[x, y)$ for the half-open interval $\{t : x \leq t < y\}$) – equivalently, the most significant decimal digit of m equals 1. OEIS A190302 (J. Krizek, May 07 2011, offset 1) [1] carries the verbatim description "*Smallest number h such that the decimal expansion of n * h starts with* 1" and defines

$$a(n) \;=\; \min\{\, h \geq 1 : \text{the decimal expansion of } n \cdot h \text{ starts with } 1 \,\},$$

with first values $1, 5, 4, 3, 2, 2, 2, 2, 2, 1, \ldots$ The entry records the companion sequence A187285(n) [2] – the least multiple of n beginning with 1 – through the quotient identity $a(n) =$ A187285$(n)/n$, and the conjecture of Felix Fröhlich (Jul 28 2018), verbatim: "*Conjecture: $a(n) < 6$ for all n (verified to $n = 10022141$).*"

Theorem 1 (exact formula)**.** *Let $n \geq 1$ and let $d \geq 0$ be the integer with $10^d \leq n < 10^{d+1}$. Then*

$$a(n) = \begin{cases} 1, & n < 2 \cdot 10^d, \\ \lceil 10^{d+1}/n \rceil, & 2 \cdot 10^d \leq n < 10^{d+1}. \end{cases}$$

Corollary 2 (Fröhlich's conjecture, OEIS A190302)**.** $1 \leq a(n) \leq 5$ *for every $n \geq 1$; in particular $a(n) < 6$.*

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2020 Mathematics Subject Classification: 11A63. Keywords: decimal expansion, leading digit, multiples, OEIS A190302, OEIS A187285, formal verification.

Corollary 3 (the five classes)**.** *For $d \geq 0$ and $h \in \{2,3,4,5\}$, an integer n with $d+1$ digits satisfies $a(n) = h$ iff*

$$\frac{10^{d+1}}{h} \;\leq\; n \;<\; \frac{10^{d+1}}{h-1};$$

and $a(n) = 1$ iff $10^d \leq n < 2 \cdot 10^d$. Each of the five values is attained (e.g. $a(2 \cdot 10^d) = 5$), so the bound 5 is sharp.

Theorem 1 and Corollary 2 are formally verified in Lean 4 with mathlib (§3). Corollary 3 is a direct consequence of the formula, stated here informally: it is not part of the Lean artifact.

Prior art. The bound $a(n) \leq 5$ was asserted *without proof* by Charles R. Greathouse IV (Mar 07 2011) in the comments of OEIS A187285 [2]: "*a(n) is in* $\{n, 2n, 3n, 4n, 5n\}$" (where $a(n)$ there denotes $\mathrm{A187285}(n) = n \cdot a(n)$); the bound itself is not claimed as new.

2 Proof of Theorem 1

Fix $n \geq 1$ and d with $10^d \leq n < 10^{d+1}$.

Case $n < 2 \cdot 10^d$ (leading digit 1). The decimal expansion of n itself starts with 1, so $h = 1$ is admissible and $a(n) = 1$.

Case $2 \cdot 10^d \leq n < 10^{d+1}$ (leading digit ≥ 2). Set $c = \lceil 10^{d+1}/n \rceil$.

Admissibility. By definition of the ceiling, $10^{d+1} \leq n \cdot c$; and $n \cdot c \leq 10^{d+1} + n - 1 < 2 \cdot 10^{d+1}$ since $n < 10^{d+1}$. Hence $n \cdot c \in [10^{d+1},\, 2 \cdot 10^{d+1})$: its decimal expansion starts with 1, and $c \geq 1$ (as $n \cdot c \geq 10^{d+1} > 0$), so c is admissible.

Minimality. Let $h \geq 1$ be admissible: $n \cdot h \in [10^k,\, 2 \cdot 10^k)$ for some k. If $k \leq d$ then, using $h \geq 1$,

$$n \cdot h \;<\; 2 \cdot 10^k \;\leq\; 2 \cdot 10^d \;\leq\; n \;\leq\; n \cdot h,$$

a contradiction. Hence $k \geq d+1$ and $n \cdot h \geq 10^{d+1}$, which gives $c = \lceil 10^{d+1}/n \rceil \leq h$.

Therefore $a(n) = c$. The bound follows: since $n \geq 2 \cdot 10^d$, $10^{d+1}/n \leq 10^{d+1}/(2 \cdot 10^d) = 5$, so $c \leq 5$; and $c \geq 2$ since $n \cdot 1 = n < 10^{d+1}$ is not admissible. □

Corollary 2 is immediate: in both cases the admissible set is nonempty, so the minimum exists, is ≥ 1, and is ≤ 5. Corollary 3 is the ceiling inversion $\lceil 10^{d+1}/n \rceil = h \iff 10^{d+1}/h \leq n < 10^{d+1}/(h-1)$; for $d \geq 1$ each interval has length $10^{d+1}/(h(h-1)) \geq 5$, hence contains an integer, and for $d = 0$ the five classes are witnessed directly by $a(1) = 1$, $a(5) = 2$, $a(4) = 3$, $a(3) = 4$ and $a(2) = 5$; so all five values occur.

3 Formal verification

The statement is formalized in Lean 4 [3] (toolchain `leanprover/lean4:v4.35.0`, mathlib `v4.35.0`, pinned manifest) in a single self-contained file `A190302.lean` (261 lines, `import Mathlib` only). The definitions read

$$\texttt{startsWithOne}(m) :\Longleftrightarrow\; \exists\, k,\; 10^k \leq m < 2 \cdot 10^k,$$

$$\texttt{a190302}(n) := \mathrm{sInf}\,\{h : \mathbb{N} \mid 1 \leq h \wedge \texttt{startsWithOne}(n \cdot h)\},$$

where sInf on $\mathbb{N}$ returns 0 on the empty set (totalizing convention; only the case $n = 0$ is affected, outside the OEIS domain). The formalized theorem is the *strengthened* statement

$$\texttt{A190302}: \quad \forall\, n \geq 1, \quad 1 \leq \texttt{a190302}(n) \ \wedge\ \texttt{a190302}(n) < 6,$$

the lower clause being necessary for faithfulness: without it, a hypothetical empty set of admissible h would satisfy $a(n) = \mathrm{sInf}\,\varnothing = 0 < 6$ vacuously. `A190302_lt_six` is the verbatim corollary $a(n) < 6$. The exact formula of Theorem 1 is carried by two declarations, each paired with a d-parametrized variant (`_of_pow`, taking d and the bounds $10^d \leq n < 10^{d+1}$ as hypotheses). The first case is `a190302_eq_one` ($a(n) = 1$ under $n < 2 \cdot 10^{\texttt{Nat.log}\,10\,n}$); the second is `a190302_eq_ceilDiv`, whose signature in the artifact is

$$\texttt{a190302_eq_ceilDiv}: \quad 2 \cdot 10^{\texttt{Nat.log}\,10\,n} \leq n \implies \texttt{a190302}\ n \ =\ 10^{(\texttt{Nat.log}\,10\,n+1)}\ \lceil/\rceil\ n,$$

where $\lceil/\rceil$ is mathlib's ceiling division (`CeilDiv.ceilDiv`: $b\ \lceil/\rceil\ a$ is the least c with $b \leq a \cdot c$, i.e. $\lceil b/a \rceil$) and `Nat.log` 10 n is the d of Theorem 1. Table 1 records the correspondence between statements in the text and declarations in the Lean artifact.

Paper item	Lean declaration	Role in formalization
"starts with 1"	`startsWithOne`	interval predicate $m \in [10^k, 2 \cdot 10^k)$
sequence $a(n)$	`a190302`	sInf of the admissible multipliers
Thm. 1, $n < 2 \cdot 10^d$	`a190302_eq_one`	exact value $a(n) = 1$, $d =$ `Nat.log` 10 n
Thm. 1, param. variant	`a190302_eq_one_of_pow`	same, d and bounds as hypotheses
Thm. 1, $n \geq 2 \cdot 10^d$	`a190302_eq_ceilDiv`	$a(n) = 10^{d+1}\lceil/\rceil n$, $d =$ `Nat.log` 10 n
Thm. 1, param. variant	`a190302_eq_ceilDiv_of_pow`	same, d and bounds as hypotheses
Cor. 2	`A190302`	strengthened statement $1 \leq a(n) < 6$
verbatim clause	`A190302_lt_six`	bare corollary $a(n) < 6$
literal reading	`startsWithOne_iff_getLastD`	interval iff leading digit 1 (`Nat.digits`)

Table 1: Correspondence between mathematical statements and Lean 4 declarations.

Faithfulness of the transcription is checked inside the artifact by two items. The declaration `startsWithOne_iff_getLastD` proves the interval predicate equivalent to the literal OEIS reading "most significant digit of m equals 1" via `Nat.digits` (the digit list is little-endian; the leading digit is its last element), and the values printed on the OEIS entry are checked by `decide` on the eleven examples $a(1)$, $a(2)$, $a(3)$, $a(4)$, $a(5)$, $a(7)$, $a(11)$, $a(12)$, $a(19)$, $a(20)$ and $a(21)$.

`lake env lean A190302.lean`: exit 0, zero `sorry`. The reports of `#print axioms` on the two named theorems, `A190302` and `A190302_lt_six`, give the three standard axioms only – `propext`, `Classical.choice`, `Quot.sound` – with **no** `native_decide` and no `sorryAx`. Independently of this report, every declaration in the file, including the finite example checks (`decide`-evaluated, hence kernel-reduced), is verified by the Lean kernel at compile time.

The declaration `A190302` was also exported together with its transitive dependency closure, and the export was replayed successfully (exit code 0) by three independent reimplementations of the Lean 4 kernel, run with the permitted-axiom list restricted to the same three axioms. What this replayed closure covers should be stated precisely: besides `A190302` itself, it contains the definitions `startsWithOne` and `a190302` and the four case declarations `a190302_eq_one`, `a190302_eq_one_of_pow`, `a190302_eq_ceilDiv` and `a190302_eq_ceilDiv_of_pow` – so the formalized content of Theorem 1 is re-verified by the external kernels as part of the closure, though

not as named export roots. Three items of the file lie outside the closure and are covered only by the in-file kernel check: `A190302_lt_six`, a downstream corollary rather than a dependency of the featured theorem; the faithfulness bridge `startsWithOne_iff_getLastD` and its auxiliary `ofDigits_dropLast_add_getLast`, which the featured theorem does not use; and the eleven anonymous spot-checks. Export files and manifests are archived with the record.

4 Remarks

- The companion sequence is thus exactly A187285(n) = n when the leading digit of n is 1, and $n \cdot \lceil 10^{d+1}/n \rceil \in [10^{d+1}, 2 \cdot 10^{d+1})$ otherwise – the least multiple of n beginning with 1, recovering Greathouse's comment A187285(n) $\in \{n, 2n, 3n, 4n, 5n\}$.
- The proof is one line in spirit – the first multiple of n at or beyond 10^{d+1} must begin with 1 – but the OEIS domain is $n \geq 1$ with no asymptotic range, so the value of the note is exactness for every n and the kernel-checked statement, not a new bound.
- In base $b \geq 2$ the same argument gives: if the leading base-b digit of n is 1 then $a_b(n) = 1$, else $a_b(n) = \lceil b^{d+1}/n \rceil \leq \lceil b/2 \rceil$ (for instance ≤ 5 in base 10, ≤ 2 in base 4; in base 2 always $a_2(n) = 1$). Remark only – not part of the Lean artifact.
- Numerical status: the conjecture was verified by Fröhlich on the entry up to $n = 10022141$; the exact formula was additionally tested against brute force on $[1, 5000]$ (total agreement; a numerical sanity check, not part of the proof).

5 Code availability

This note, the Lean file `A190302.lean`, the pinned build files and the companion verification archive are permanently archived together on Zenodo [4] at `https://doi.org/10.5281/zenodo.23957595` (concept DOI, always resolving to the latest version), under the Apache-2.0 license.

Use of AI tools

Generative AI tools assisted in independent re-transcription of the OEIS entry, adversarial auditing, proof sketching, and text review, within a verification methodology designed and supervised by the author. All theorems are verified from standard axioms only in Lean 4, with the exported featured statement replayed by three independent external type-checkers and the exports archived. The author selected the problems and proof strategies, arbitrated all discrepancies, verified OEIS fidelity, and takes full responsibility for this work.

Funding

No funding was received for this work.

Competing interests

The author declares none.

References

[1] J. Krizek, OEIS A190302 (May 07 2011); conjecture comment F. Fröhlich, Jul 28 2018. *The On-Line Encyclopedia of Integer Sequences*, N. J. A. Sloane ed., `https://oeis.org/A190302`.

[2] OEIS A187285 (least multiple of n beginning with 1); comment C. R. Greathouse IV, Mar 07 2011: "a(n) is in $\{n, 2n, 3n, 4n, 5n\}$", `https://oeis.org/A187285`.

[3] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Comput. Sci., vol. 12699, Springer, 2021, pp. 625–635; The mathlib Community, The Lean mathematical library, in *Proc. 9th ACM SIGPLAN Int. Conf. on Certified Programs and Proofs (CPP 2020)*, ACM, 2020, pp. 367–381 (toolchain `leanprover/lean4:v4.35.0`, mathlib `v4.35.0`).

[4] Frédéric Gaudin, *Smallest multipliers with leading digit one and a conjecture of Fröhlich (OEIS A190302)* - Zenodo deposit of this note, the Lean file and the verification archive, `https://doi.org/10.5281/zenodo.23957595` (concept DOI).